Two goals at once

Optimization · Level 4 · Lesson 1



Today we will

- Cross out every plan another plan beats: no worse on both goals, better on one.
- Choose among the rest with weights we agree first.
- Say what one more unit of a limit is worth.

Remember

- A beaten plan is never right: keep the Pareto front.
- Weights are values: agree them openly, before seeing the winner.
- Solve again with one more unit to find what a limit is worth.

Beaten or not?

1. Warm-up (5 min): Shorter or gentler sports day?
2. Teach (8 min): Trade-offs and plans nobody beats.
3. Cross out (10 min): Groups remove the beaten plans.
4. Weigh (10 min): Agree weights, then find the winner.
5. Wrap-up (7 min): How did the weights change the choice?

Write the four plans on cards, one set per group. Graph paper helps: put hours across and waiting pupils up. A beaten plan sits straight above the plan that beats it, straight to its right, or above and to the right: R sits straight above Q (same hours, more pupils waiting).

Let’s discuss

1. Why can a beaten plan never be the right choice?
2. Who should agree the weights for an exam week, and when?
3. Where would one more hour or one more coin help our school most?

Exit ticket

- Name one plan you crossed out, the plan that beats it, and why.